

Task 19 — Application Model Serializing

AI / ML Developer · Phase 1 Industry Immersion · Task Brief

AI / ML DEVELOPER

PlaceMux · Phase 1 · Task Brief

DIFFICULTY

Mastery · Expert

Level 19 of 20 · ramp 90%

Objective

What this task is: Serialise the trained model and its preprocessing so it can be loaded and served reliably in an application.

Why it matters: A model stuck in a notebook helps no one. Serialisation is the hand-off from research to a deployable artifact.

What you will deliver

- A serialised, versioned model+preprocessor artifact with metadata and a tested load/predict path.

Inputs & prerequisites

- A reproducible ML environment with fixed seeds.
- Train/validation/test discipline and a baseline-first mindset.
- Honest evaluation: the right metric, on unseen data.
- Required tools for this task: joblib/pickle, a metadata JSON, FastAPI predict stub, pydantic

Step-by-step execution

Work through these in order. Don't skip the verification steps — that's where 'looks done' becomes 'is done'.

1. Bundle the fitted pipeline (preprocess + model) into one artifact.
2. Save it with joblib/pickle plus a metadata file.
3. Record library versions and training metrics.
4. Write a load-and-predict function with input validation.
5. Test loading in a fresh environment and predicting.
6. Version the artifact for traceability.

Tools & stack

- joblib/pickle
- a metadata JSON
- FastAPI predict stub
- pydantic

Definition of Done

You are done when all of these are true and demoable

- A serialised, versioned model+preprocessor artifact with metadata and a tested load/predict path.
- Demonstrable live on real (even if small) data, not just described.

Pitfalls to avoid

Worry if any of these are true of your work

- Saving model but not preprocessor.
- Version mismatch breaking load.
- No metadata/lineage.

Where this fits in the track

Over Phase 1 you go from a clean environment to a deployed, callable model. You learn to build models you can measure, explain and serve -- not just ones that run.